SAI SANTOSH CHEETHIRALA

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Systems engineer who builds storage and replication, then proves where it breaks. Found 56 acknowledged writes lost in a shipping distributed database at a durability level whose gap was documented but never demonstrated, and built a sharded, Raft-replicated key/value store in Rust with a from-scratch LSM engine.

EXPERIENCE

PESU Intelligence Labs — Research Intern, RAG & LLM Reliability **Bengaluru | Jun 2026 – Jul 2026**

No Single Culprit — Causal Attribution Benchmark & Evaluation Harness for Multi-Hop RAG **github.com/santoshcheethiralame-dot/LINEUP**

Python, pytest, GitHub Actions (CI/CD), REST API, Streamlit, vLLM, BM25, PyTorch, HuggingFace Transformers, Conformal Prediction

- Built and shipped a 7.2K-LOC evaluation service — public REST API, pooled leaderboard, and Streamlit explorer — with 135 automated tests gating every change on GitHub Actions CI
- Ran ~100K teacher-forced inferences across open-weight models (7B-120B) served with vLLM against a live BM25 retriever, sweeping three multi-hop datasets and retrieval depths k=20-50
- Replaced single-passage attribution with a calibrated set-valued method, lifting top-1 accuracy from 0.35 to 0.64 at 94% conformal coverage; first-author paper submitting to TMLR

PROJECTS

highwater — Jepsen-Style Durability Analysis of Couchbase **github.com/santoshcheethiralame-dot/HighWater**

Go, Couchbase DCP, Porcupine linearizability checker, gocbcore patch, fault injection

- Ran a Jepsen-style durability analysis of Couchbase — one of the few major distributed databases with no published Jepsen report — showing 56 acknowledged writes lost (50 regressed, 6 absent) at durability level "majority" when the in-memory majority is killed before the disk flush completes, with the history proven non-linearizable
- Built a white-box harness that taps the DCP stream to observe prepare/commit/abort directly, rather than inferring from client responses, so "never prepared" and "prepared then aborted" are distinguishable; required patching gocbcore to expose the stream
- Confirmed the result under two independent methods that agree on both arms — a direct value comparison and Porcupine, a general linearizability checker with no knowledge of durability levels — and showed the control arm (persistToMajority) loses zero writes and stays linearizable

tessera — Distributed, Sharded, Replicated Key/Value Store (Rust, std-only) **github.com/santoshcheethiralame-dot/TESSERA**

Rust (0 external crates), Raft, LSM-tree, Deterministic Simulation Testing, Linearizability Checking, MVCC

- Built a distributed, sharded, replicated key/value store in pure Rust with zero external crates: a from-scratch LSM storage engine (memtable, SSTables, write-ahead log, size-tiered compaction, MVCC) and per-shard Raft replication behind a coordinator and routing client (CockroachDB/TiKV shape, scaled down)
- Implemented snapshots and log compaction, linearizable reads via leader leases, joint-consensus membership changes, and crash-safe persistence (term/vote/log fsync'd, temp-file + atomic rename); benchmarked p50/p99 latency and throughput on real disk over a real TCP cluster
- Caught a real split-brain bug via seed 99 — a duplicated vote reply let a candidate win a sub-majority, producing two leaders in one term — under deterministic simulation testing that replays any failure exactly from its seed

GlassBox — Full-Stack Model Interpretability Platform (Deployed) **numero00-glassbox.hf.space**

Python, FastAPI, React, TypeScript, Docker, REST APIs, PyTorch, TransformerLens, SAELens

- Ran a 2.6B model on a free tier by diagnosing the load as host-RAM-bound rather than VRAM-bound and re-architecting with CPU-staged low-memory weight conversion to hold peak under a 12 GB ceiling

TECHNICAL SKILLS

Systems & Backend: Distributed systems (Raft, LSM storage, MVCC), FastAPI, REST APIs, Docker, Linux, WebGPU/WGSL, vLLM

Languages: Python, C, Rust, TypeScript, JavaScript, SQL, WGSL

Testing & Reliability: Deterministic simulation testing, randomized fault injection, linearizability checking, regression testing, pytest, CI/CD (GitHub Actions), model evaluation

Tools: Git, GitHub Actions, Linux, Hugging Face Spaces, GPU compute, numerical gradient checking

EDUCATION

PES University — B.Tech, Computer Science Engineering **Bengaluru | Sep 2024 – May 2028 (Expected)**

CGPA 7.95/10 (through Sem 4) — CNR Merit Scholarship (top 25% cohort), Sem 1 — Academic Distinction Award (SGPA 7.75+), Sem 2 & Sem 4

Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Operating Systems, Computer Networks, Machine Learning, Deep Learning, NLP, Linear Algebra, Data Science

LEADERSHIP & ACTIVITIES

Sahay AI — Student AI Club, PES University — Multimedia Domain Head **Oct 2025 – Mar 2026**

- Built a content pipeline and posting schedule for a 50+ member AI club, driving 5-6x growth in content views over three months
- National-Level Air Rifle Competitor - 10m Air Rifle, National School Games (a precision event built on sustained focus)